Predicting Asteroid Diameter Based on Photometric Parameters and Orbital Characteristics

Micro-Project 4

GitHub Repository

https://github.com/Ethereal-Mining-Company/736D616C6C2D626F64792D7633

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October 11, 2026

- DISCLAIMER -

*The term **XGBOOST** is used to refer to **Gradient Boosting Regression (GBR)** throughout the framework. This is to limit code refactoring and re-running training to replace XGBOOST with GBR.*

Problem Statement

Small-body diameter prediction methods:

(1) A combination of very high data collection cost, very high predictive power, and low observational reliability features like albedo (Group 3) with low data collection cost, medium observational reliability features, like absolute magnitude (H) (Group 2)

(2) A combination of low data collection cost, high observational reliability features (Group 1) with low data collection cost, medium observational reliability features, like absolute magnitude (H) (Group 2)(Ge, 2026; Murcia, 2023; Robinson, 2024).

The project looks to determine if the diameter of a small-body can accurately be predicted with only Group 1 features to maintain the low-cost, high-reliability of features that are inexpensive and not variable or prone to error while observing.

Hypothesis Formulation

The project utilizes a two-model comparison framework that will be assessed from a statistical and operational perspective. The regression models target diameter (Y) using two feature subsets: an expanded set of low-cost, high-reliability observations with low-cost, medium-reliability observations ($\mathbf{X}_{full}$) and a restricted set of only 'low-cost', 'high reliability' observations ($\mathbf{X}_{restricted}$).

The restricted model ($\mathbf{X}_{restricted}$) is preferred due to the lower number and higher realiability of features. Current peer-reviewed research dictates that absolute magnitude (H), a Group 2 feature, is mathematically required to calculate diameter (Ge, 2026).

The hypothesis determines if small-body diameter (Y) can be accurately predicted within a statistical and operational threshold () by the restrictive feature subset compared to the full feature subset, quantified as:

$$\Delta R^2_{test} = R^2_{Model(\mathbf{X}full)} - R^2_{Model(\mathbf{X}restricted)}$$

Hypothesis Formulation

Statistical Hypothesis
The statistical hypothesis tests if the restrictive feature subset of Group 1 features maintains a statistically acceptable degree of accuracy in predicting target diameter (Y) relative to the expanded feature subset of Group 1 and Group 2 features.

- **Statistical Null Hypothesis ($H_{0,stat}$):** $\Delta R^2_{test} \leq 0$
  The inclusion of Group 2 features yield no improvement or a decrease in the out-of-sample ΔR^2_{test}, indicating that the expanded features **do not** increase the accuracy of predicting the target diameter (Y).
- **Statistical Alternative Hypothesis ($H_{1,stat}$):** $\Delta R^2_{test} > 0$
  The inclusion of Group 2 features yields an increase in the out-of-sample ΔR^2_{test}, indicating that the expanded features **do** increase the accuracy of predicting the target diameter (Y).

Hypothesis Formulation

Operational Hypothesis

The operational hypothesis tests if the restrictive feature subset of Group 1 features maintains an operationally and practically acceptable degree of accuracy () in predicting target diameter (Y) relative to the expanded feature subset of Group 1 and Group 2 features.

- **Operational Null Hypothesis ($H_{0,ops}$):** $\Delta R^2_{test} > \delta$
  The inclusion of Group 2 features yield an operationally significant increase in accuracy to predict target diameter (Y), indicating that the restricted feature subset **is not** sufficient enough for operational use.
- **Operational Alternative Hypothesis ($H_{1,ops}$):** $\Delta R^2_{test} \leq \delta$
  The inclusion of Group 2 features does not yield an operationally significant increase in accuracy to predict target diameter (Y), indicating that the restricted feature subset **is** sufficient enough for operational use

Acquire

The JPL NASA Small-Body Dataset was chosen for the following reasons:

- Contains Group 1 and Group 2 characteristics such as absolute magnitude *(H)*, semi-major axis *(a)*, eccentricity *(e)*, and inclination *(i)* which directly relate to the problem statement and hypothesis requirements necessary for the expanded features subset and restricted feature subset.
- NASA is an authoritative, reliable, and primary repository of small-body data that is also open-source and available to the public allowing reproduction of testing for peer-review.

The JPL NASA Small-Body Dataset was queried to meet the target variable diameter *(Y)* to be used in out-of-sample testing. The query exclusively included asteroids with a known diameter greater than zero kilometers (0 km).

Prepare

Data has been prepared for regression analysis in the following ways:

1. **Data Structure Integrity:** Validate all data types are correct.
2. **Data Domain Validation:** Remove data is implausible such as negative diameter.
3. **Duplicate Data Removal:** Remove duplicate rows of data and observations of the same small-body.
4. **Target Variable Isolation (diameter):** Diameter (Y) is the target variable and is required for training.
5. **Exclusion of High-Cost Features (albedo):** Group 3 characteristic are explicitly dropped from the dataset.
6. **Feature Subset Selection:** The model requires the target diameter and features that fall into Group 1.
7. **Feature Completeness:** Entries with missing values in the selected feature subsets were removed.
8. **Target Variable Distribution Scaling:** Scale diameter *(Y)* using a base-10 logarithmic formula.
9. **Outlier Detection:** Represent real small-bodies and may not be removed.
10. **Feature Independence:** Remove certain highly correlated features.

Prepare

1. Data Structure Integrity

All features are the correct data type. Discrete values like `skpid` (the small-body unique identifier) and `condition code` are integers (*`int`*) and all numerical measurements and values like `a`, `e`, and `i` are floats (*`float64`*).

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 139582 entries, 0 to 139581
Data columns (total 18 columns):
 #   Column          Non-Null Count   Dtype  
---  ------          --------------   -----  
 0   spkid           139582 non-null  int64  
 1   diameter        139582 non-null  float64
 2   albedo          138459 non-null  float64
 3   H               139203 non-null  float64
 4   full_name       139582 non-null  str    
 5   a               139582 non-null  float64
 6   e               139582 non-null  float64
 7   i               139582 non-null  float64
 8   om              139582 non-null  float64
 9   w               139582 non-null  float64
 10  q               139582 non-null  float64
 11  ad              139582 non-null  float64
 12  per_y           139582 non-null  float64
 13  data_arc        139577 non-null  float64
 14  condition_code  139582 non-null  int64  
 15  n_obs_used      139582 non-null  int64  
 16  n_del_obs_used  243 non-null     float64
 17  n_dop_obs_used  243 non-null     float64
dtypes: float64(14), int64(3), str(1)
memory usage: 21.7 MB
```

Prepare

2. Data Domain Validation

Removed data that is implausible. Since the data is not validated by an astrophysicist, the project will be limited to checking just the `diameter` (`Y`). Diameters are all valid because `(df['diameter'] <= 0).sum() = 0`.

```
print(f"Number of invalid diameter records: {(df['diameter'] <= 0).sum()}")
```

```
Number of invalid diameter records: 0
```

3. Duplicate Data Removal

Removed rows with identical values in every column and where rows share the same `skpid`. Confirmed removed when `df.duplicated().sum() = 0` and `df.duplicated(subset=['spkid']).sum() = 0`.

```
print("# of rows with identical values in every single column: ", df.duplicated().sum())
print("# of rows with the same skpid", df.duplicated(subset=['spkid']).sum())
```

```
# of rows with identical values in every single column:  0
# of rows with the same skpid 0
```

4. Target Variable Isolation

All entries without a `diameter` (`Y`) have been removed. Confirmed because `.isnull().sum() = 0` for `diameter` (`Y`).

Prepare

5. Exclusion of High-Cost Features

Drop `albedo` because it is a high-cost observation that will not be included in the model. Confirmed removed because feature does not show up in `.info()` or `.isnull().sum()`.

```
df_clean = df.dropna(subset=y).drop(columns=['albedo'])
df_clean.isnull().sum()
```

```
spkid                0
diameter             0
H                  379
full_name            0
a                    0
e                    0
i                    0
om                   0
w                    0
q                    0
ad                   0
per_y                0
data_arc             5
condition_code       0
n_obs_used           0
n_del_obs_used    139339
n_dop_obs_used    139339
dtype: int64
```

Prepare

6. Feature Subset Selection

Removed all features that are not part of the full feature subset because they are metadata or not orbital characteristics used to determine `diameter` (`Y`).

```
df_clean = df_clean[[y] + x_full + ['ad', 'q', 'om', 'w']]
df_clean.head()
```

	diameter	a	e	i	H	ad	q	om	w
0	939.400	2.766	0.0797	10.59	3.34	2.99	2.545	80.25	73.29
1	513.000	2.770	0.2307	34.93	4.12	3.41	2.131	172.89	310.97
2	246.596	2.671	0.2557	12.99	5.19	3.35	1.988	169.81	247.90
3	522.770	2.361	0.0902	7.14	3.25	2.57	2.148	103.70	151.47
4	106.699	2.577	0.1876	5.36	6.95	3.06	2.093	141.45	359.38

Prepare

7. Feature Completeness

Checked for any missing or null values using using `.isna().sum()`. If missing values, removed the entire entry using `.dropna()`. Since the only feature with missing values is `H` the entries can be deleted without review. This is because `H` is the feature difference in the hypothesis and is fundamental and required to test the model. For the same reasons, it is not appropriate to estimate this value.

```
df_clean.isna().sum()
```

```
diameter      0
a             0
e             0
i             0
H           379
ad            0
q             0
om            0
w             0
dtype: int64
```

```
df_clean.dropna(subset=['H'], inplace=True)
df_clean.isna().sum()
```

```
diameter    0
a           0
e           0
i           0
H           0
ad          0
q           0
om          0
w           0
dtype: int64
```

Prepare

8. Target Variable Distribution Scaling

Verified `target` `diameter` skew is greater than one (`1`). Since it was highly skewed, the `diameter` (`Y`) was transformed into a `log` `base-10` scale, `diameter_log` , so the regression model does not improperly weight errors between small-body sizes. After the transformation, `diameter_log` was checked for errors in values and the entries were removed as necessary.

```
df_clean['diameter'].describe()
```

```
count    139203.000000
mean          5.469118
std           9.318406
min           0.002500
25%           2.773000
50%           3.957000
75%           5.739000
max         939.400000
Name: diameter, dtype: float64
```

```
print(df_clean['diameter'].skew())
```

```
25.930498594815756
```

```
df_clean['diameter_log'] = np.log10(df_clean['diameter'])
print(f"Total infinity anomalies found: {np.isinf(df_clean['diameter_log']).sum()}")
print(f"Total na values found: {df_clean['diameter_log'].isna().sum()}")
df_clean['diameter_log'].describe()
```

```
Total infinity anomalies found: 0
Total na values found: 0
count    139203.000000
mean          0.615454
std           0.282694
min          -2.602060
25%           0.442950
50%           0.597366
75%           0.758836
max           2.972851
Name: diameter_log, dtype: float64
```

Prepare

9. Outlier Detection

Calculated the lower and upper bounds from the `interquartile range` (`IQR`) and reviewed all outliers. No entries were determiend to be errors so outliers were kept in the dataset because they represent a real small-body that is part of the population of data. The `diameter` was reviewed to compare to `diameter_log` but will not be used in the model.

```
q1 = df_clean.quantile(0.25)
q3 = df_clean.quantile(0.75)
iqr = q3 - q1

lower_bound = q1 - 1.5 * iqr
upper_bound = q3 + 1.5 * iqr

total_rows = len(df_clean)

for col in df_clean.select_dtypes(include=['number']).columns:
    low_count = (df_clean[col] < lower_bound[col]).sum()
    high_count = (df_clean[col] > upper_bound[col]).sum()

    low_pct = (low_count / total_rows) * 100
    high_pct = (high_count / total_rows) * 100

    print(f"{col} -> Lower Outliers: {low_count} ({low_pct:.2f}%) | Upper Outliers: {high_count} ({high_pct:.2f}%)")
```

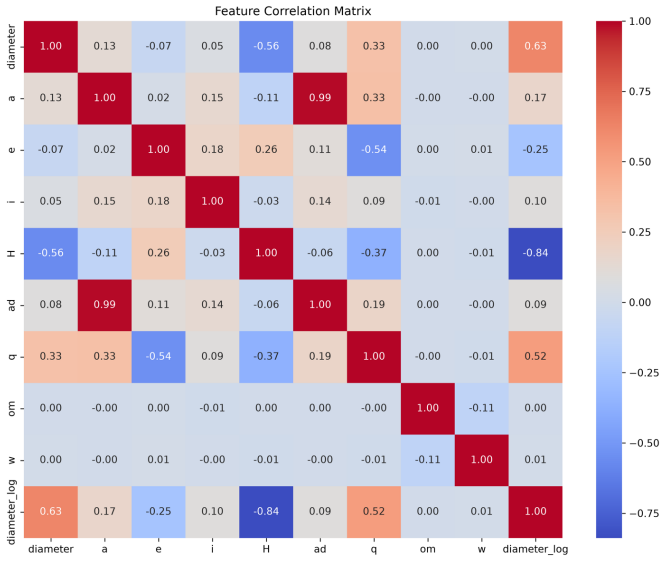
```
diameter -> Lower Outliers: 0 (0.00%) | Upper Outliers: 9809 (7.05%)
a -> Lower Outliers: 579 (0.42%) | Upper Outliers: 3060 (2.20%)
e -> Lower Outliers: 0 (0.00%) | Upper Outliers: 1886 (1.35%)
i -> Lower Outliers: 0 (0.00%) | Upper Outliers: 4322 (3.10%)
H -> Lower Outliers: 3075 (2.21%) | Upper Outliers: 815 (0.59%)
ad -> Lower Outliers: 449 (0.32%) | Upper Outliers: 3365 (2.42%)
q -> Lower Outliers: 1091 (0.78%) | Upper Outliers: 1983 (1.42%)
om -> Lower Outliers: 0 (0.00%) | Upper Outliers: 0 (0.00%)
w -> Lower Outliers: 0 (0.00%) | Upper Outliers: 0 (0.00%)
diameter_log -> Lower Outliers: 840 (0.60%) | Upper Outliers: 3510 (2.52%)
```

Prepare

10. Feature Independence

Created a `heatmap` to review highly correlated features. `a` and `ad` are highly correlated. Since `ad` is derived from `a` as `ad = a x (1 + e)` it will be removed so it does not get improperly weighted in the regression analysis. Features `om` and `w` have no feature-to-feature correlation or feature-to-target correlation. These will be kept since tree-based regression can factor these out if they are unnecessary.

```
# create correlation matrix to see if any variables inherently correlate to diameter
plt.figure(figsize=(10,8))
sns.heatmap(df_clean.corr(numeric_only=True), annot=True, cmap='coolwarm', fmt=".2f")
plt.title("Feature Correlation Matrix")
plt.tight_layout()
plt.savefig("charts/feature_correlation_matrix.png", dpi=300)
plt.show()
```



Analyze (EDA)

Data has been analyzed in the following ways:

1. **Individual Feature Boxplots:** Visualize distribution of individual features.
2. **Target Normalization Check:** Visualize log transformation of target.
3. **Primary Signal Feature:** Visualize absolute magnitude (H) versus target.
4. **Orbital Characteristic Features:** Visualize features.
5. **Excluded Features:** Determine features outside of analysis scope.

Analyze (EDA)

1. Individual Feature Boxplots

Visually reviewed distribution of individual features. Several of the physical features (`a` , `e` , and `i`) are heavily skewed. This indicates `linear regression models` are not ideal and supports utilizing `tree-based models` . `e` (`eccentricity`) represents the orbit of a small-body with `Earth` having an `e ≈ 0.017` . This means the small-body data set has a relative stable orbit similar to `Earth` while some outliers have less stable orbits. `i` represents `inclination` relative to `Earth's` (`Earth's i = 0°`). This means the small-body dataset distribution of `i` has a similar inclination as `Earth` which makes sense given they are under the same orbital parameters as `Earth` . `a` represents the `semi-major axis` which is the average distance from the Sun. The distribution of `a` means that most of the small-bodies are in the main asteroid belt between Mars and Jupiter with some outliers as far out as Neptune or farther.

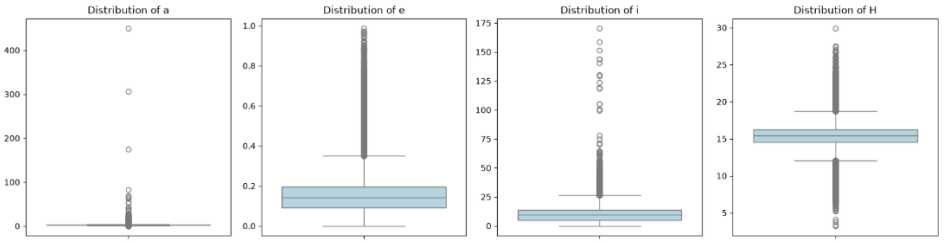
All features have acceptable distributions in context to small-bodies.

```
fig, axes = plt.subplots(2, 4, figsize=(16, 8))
axes = axes.flatten()

for idx, col in enumerate(x_full):
    sns.boxplot(data=df_clean, y=col, ax=axes[idx], color='lightblue')
    axes[idx].set_title(f'Distribution of {col}')
    axes[idx].set_ylabel('')

for i in range(len(x_full), len(axes)):
    fig.delaxes(axes[i])

plt.tight_layout()
plt.savefig("charts/feature_distros.png", dpi=300)
plt.show()
```



Analyze (EDA)

2. Target Normalization Check

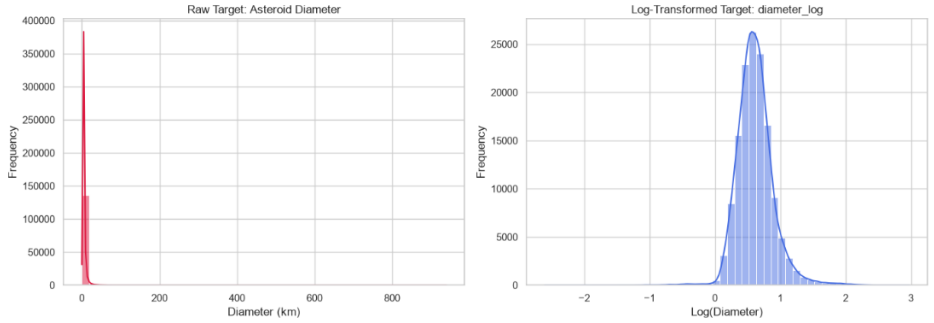
Compared the `diameter` to `diameter_log` to visualize skew. The raw `target` `diameter` is `power-law` skewed to the right. The `logarithmic` transformation decompresses the skew and better prepares the data for the `tree-based model` analysis.

```
sns.set_theme(style="whitegrid")
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

sns.histplot(df_clean['diameter'], bins=50, kde=True, ax=axes[0], color='crimson')
axes[0].set_title('Raw Target: Asteroid Diameter')
axes[0].set_xlabel('Diameter (km)')
axes[0].set_ylabel('Frequency')

sns.histplot(df_clean['diameter_log'], bins=50, kde=True, ax=axes[1], color='royalblue')
axes[1].set_title('Log-Transformed Target: diameter_log')
axes[1].set_xlabel('Log(Diameter)')
axes[1].set_ylabel('Frequency')

plt.tight_layout()
plt.savefig("charts/target_log_transform.png", dpi=300)
plt.show()
```



Analyze (EDA)

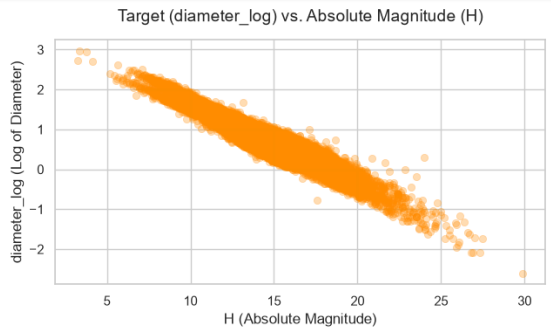
3. Primary Signal Feature

`H` is the signal feature that differentiates the full and restricted feature subsets and is highly correlated to `diameter`. The scatterplot visually confirms this correlation and shows an inverse relationship between `H` and `diameter_log`. `Absolute magnitude (H)` represents `intrinsic brightness` of a small-body where a smaller `H` value means higher brightness. This trend shows that a small-body with a higher brightness (lower `H` value) tends to be larger and a small-body with a lower brightness (higher `H` value) tends to be smaller. This visualization supports the `peer-reviewed` research that `H` is highly correlated to `diameter` and important to predicting a small-body `diameter`.

```
sns.set_theme(style="whitegrid")
plt.figure(figsize=(6.5, 4))

sns.scatterplot(data=df_clean, x='H', y='diameter_log', alpha=0.3, color='darkorange', edgecolor=None)
plt.title('Target (diameter_log) vs. Absolute Magnitude (H)', fontsize=14, pad=15)
plt.xlabel('H (Absolute Magnitude)', fontsize=12)
plt.ylabel('diameter_log (Log of Diameter)', fontsize=12)

plt.tight_layout()
plt.savefig("charts/target_v_abs_mag", dpi=300)
plt.show()
```

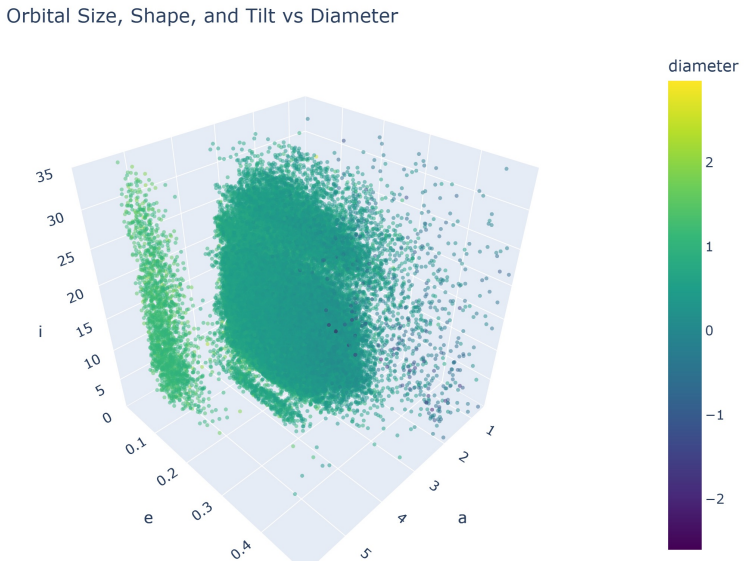


Analyze (EDA)

4. Orbital Characteristic Features

Fundamental Orbital Elements

Visualizing `a`, `e`, and `i` (independent orbital characteristics) vs. `diameter` to see if there is a correlation between these three features and `diameter`. The visualization reveals distinct clustering demonstrating that certain orbital groupings share diameter profiles. This provides reasonable evidence to use these features as candidates for out-of-sample testing in the model.



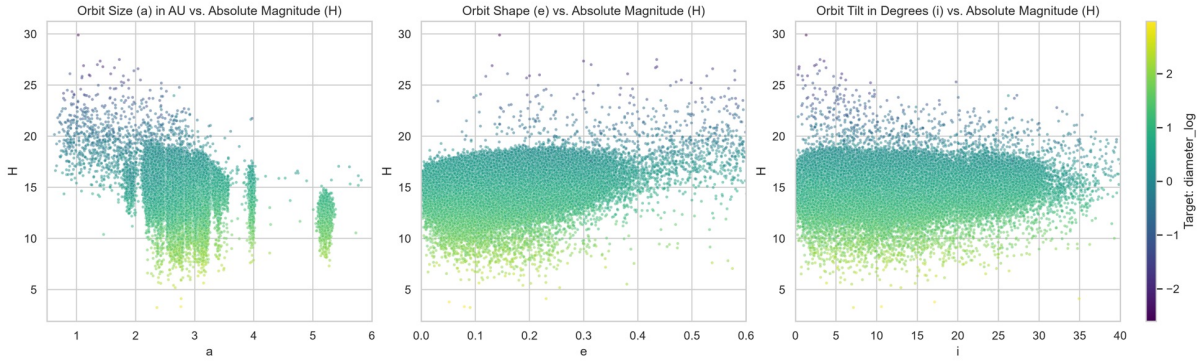
Analyze (EDA)

4. Orbital Characteristic Features

Absolute Magnitude (H) and Restricted Subset Fundamental Characteristics

Absolute Magnitude (H) versus Individual Restricted Subset Fundamental Characteristics

Visualize a, e, and i individually vs. H in terms of diameter_log. Visualization shows a clear indication that H is a primary indicator for predicting diameter but no distinct correlation exists between H and any of the individual features.



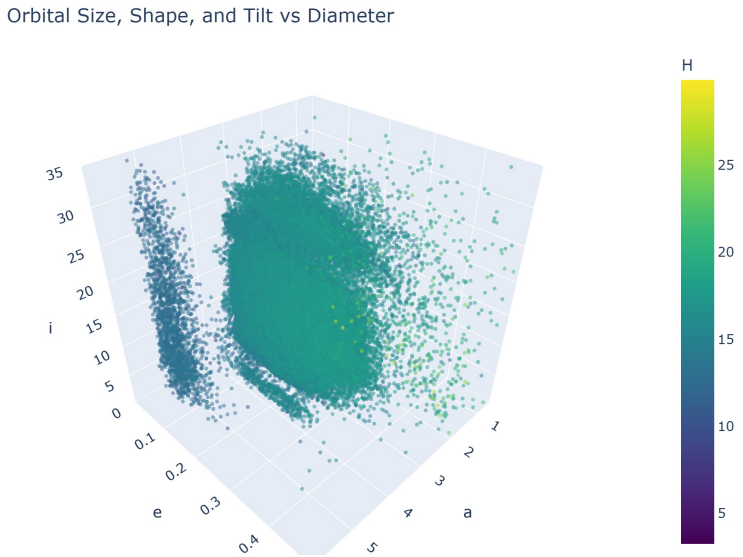
Analyze (EDA)

4. Orbital Characteristic Features

Absolute Magnitude (H) and Restricted Subset Fundamental Characteristics

Absolute Magnitude (H) in Relation to Restricted Subset Fundamental Characteristics

Visualizing a, e, and i (independent orbital characteristics) vs. absolute magnitude (H) to see if there is a correlation between these three features and H. The visualization reveals distinct clustering demonstrating that certain orbital groupings share magnitude profiles. This makes sense since H and diameter are closely related so the independent orbital characteristics should exhibit a nearly identical structural trend when mapped by H as they do when mapped by diameter.



Analyze (EDA)

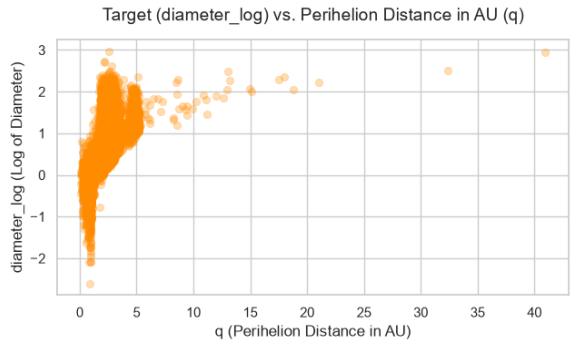
Derived Orbital Characteristics

`q` is a derived characteristic from `a` and `e` as `q = a(1-e)`. Visualizing `q` vs `diameter_log` shows a trend that the farther from the Sun a small-body gets, the larger it gets. IT has a strong `linear slope` specifically at `1 AU` up to `6 AU`. Logically this does not make sense since the size of a small-body is not dictated by the distance from the Sun. There are small and large small-bodies near and far from the Sun. Since this is a `bias` in interpretation and not a flaw in the data, `q` can be kept in the dataset similar to `om` and `w` to be included or excluded in the final model determined at a later step.

```
sns.set_theme(style="whitegrid")
plt.figure(figsize=(6.5, 4))

sns.scatterplot(data=df_clean, x='q', y='diameter_log', alpha=0.3, color='darkorange', edgecolor=None)
plt.title('Target (diameter_log) vs. Perihelion Distance in AU (q)', fontsize=14, pad=15)
plt.xlabel('q (Perihelion Distance in AU)', fontsize=12)
plt.ylabel('diameter_log (Log of Diameter)', fontsize=12)
plt.tight_layout()

plt.savefig("charts/diameter_log_v_peri_dist.png", dpi=300)
plt.show()
```

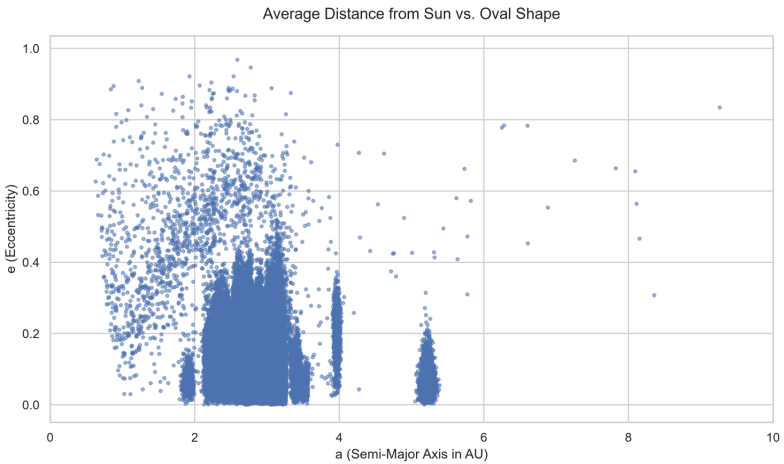


Analyze (EDA)

4. Orbital Characteristic Features

Semi-Major Axis v. Eccentricity

Visualizing a and e shows us the relationship between the average distance from the Sun and the orbital shape. The visualization does not show a relationship between a and e because there is not a linear relationship. The visualization does show that small-bodies tend to hold specific orbital distances with a major segment between 2 and 3.5 AU and another smaller segment around 5.2 AU. This is consistent with the main asteroid belt between Mars and Jupiter and a smaller one at Jupiter's orbit.

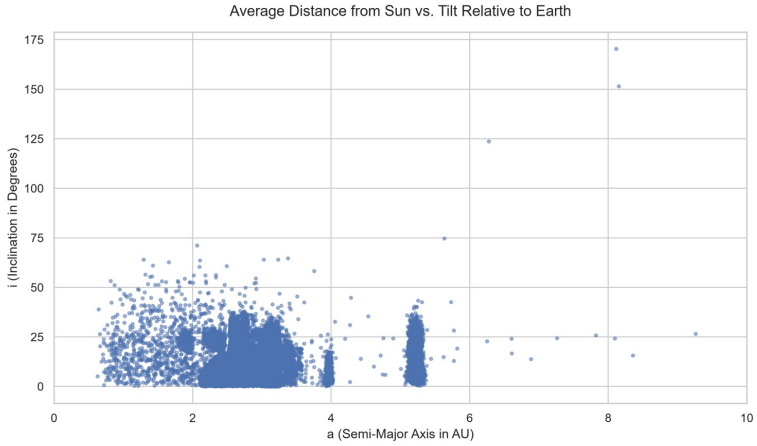


Analyze (EDA)

4. Orbital Characteristic Features

Semi-Major Axis v. Inclination

Visualizing a and i shows us the relationship between the average distance from the Sun and the inclination relative to Earth. The visualization does not show a relationship between a and i because there is not a linear relationship. The visualization does show that small-bodies throughout the inner solar system tend to have the same inclination as Earth.

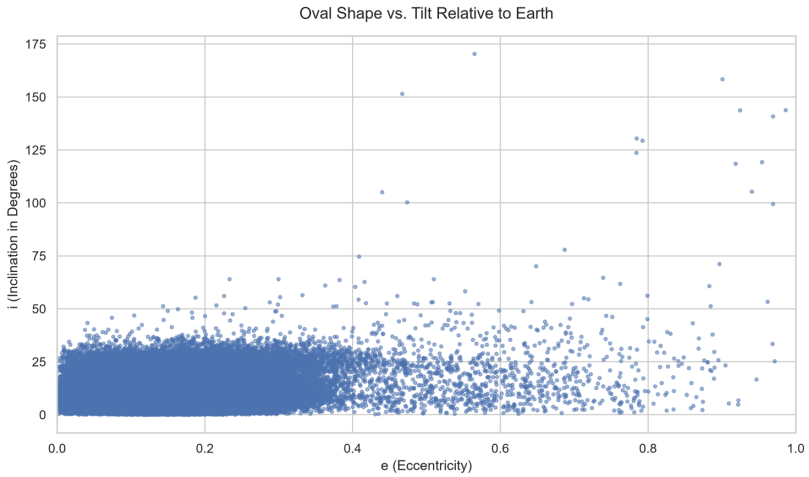


Analyze (EDA)

4. Orbital Characteristic Features

Eccentricity v. Inclination

Visualizing e and i shows us the relationship between the orbital shape and the elliptical plane. The visualization does not show a correlation between e and i because there is not a linear relationship.



Analyze (EDA)

5. Excluded Features

om and w were excluded from visualization as the feature correlation matrix determined there was not a strong correlation between these features and other features or the target. While these features may or may not introduce noise to the tree-based regression models, peer-reviewed analysis has determined that they do not provide strong predictive indicators for diameter. Due to the computing power and time to analyze additional features in the dataset, these features will only analyzed as an ad hoc if time permits prior to the project deadline.

Analyze (EDA)

1. EDA Insights

Preliminary analysis of the dataset shows that the key indicators of a small-body size (target diameter (Y)) are absolute magnitude (H), semi-major axis (a), eccentricity (e), and inclination (i). There is a strong non-linear relationship between diameter and H indicating that H is a strong predictor of diameter. While there were no linear correlations between individual a, e, i features and diameter or H, comparison of a, e, and i mapped to diameter and a, e, and i mapped to H provided similar results. This provides strong evidence that a, e, and i could meet the operational threshold to predict diameter without the need for H. The EDA alone is not enough to conclude if the null hypothesis or the alternative hypothesis are correct, but there is a pattern and relationship between a, e, i, and diameter.

Analyze (EDA)

2. Feature Determination

Analysis of the features indicates that a, e, and i are strong candidates for the regression model because they have a non-linear, structural relationship to the target. The feature q is reserved for future determination for inclusion or exclusion in the regression model since it is derived from a and e which could introduce multicollinearity or noise. Features om and w can be excluded. While om and w are not derived from any other features, the lack of correlation to the target may introduce noise to the regression model that could lead to overfitting.

Analyze (EDA)

3. Model Selection

An analysis of the dataset and feature determination support using a random forest regression model to test the hypothesis. Small-bodies form dense structural clustering in the feature space and a highly skewed target distribution. Small-bodies also can be observed at great distances which creates mathematical outliers. Physical characteristics and derived characteristics may have no correlation to the size of a small-body or introduce multicollinearity. A random forest regression model uses piecewise step-function approximations to handle extreme outliers and noise of derived or irrelevant features. This is better than linear regression or other regression models that try to fit data into a straight line or curve. This will result in stronger models to increase accuracy to predict small-body sizes and to test the hypothesis.

Analyze (Models)

Ran six regression models on the cleaned dataset. These models were chosen because they can best handle the non-linear, heavily skewed data of asteroids. The two classical machine learning models are Random Forest (RF) and Gradient Boosting Regression (GBR, XGBoost in code) which utilize decision-trees to ask questions to split data into distinct numerical groups to calculate a continuous prediction value.

The four deep learning models are Wide & Deep Network (W&D), Tabular ResNet (TabResNet), FT-Transformer and Gated ResNet (GRN) used because they handle tabular data like in the structured dataset. W&D is used because the wide path can learn the known mathematical rules like the correlation of absolute magnitude (H) and target diameter while the deep path can learn any subtle patterns with the orbital characteristics (a, e, i). TabResNet minimizes the effects of vanishing gradient problem as the network gets deeper. FT-Transformer was selected to map the complex dependencies of the features together. It looks at them as a unified system rather than isolated variables. GRN was selected because it is similar to TabResNet but will use gates to filter out noisy data to help predict the patterns in orbital characteristics, specifically in the restricted subset.

Each model was trained with the full and restricted subsets and the results logged for analysis. Training utilized a five-fold cross validation where each fold and fold results were logged as well.

Analyze (Models)

```
================================================
INITIALIZING RANDOM FOREST PIPELINE
================================================
Dataset Path:     data/cleaned-dataset-jpl-small-body-asteroids.csv
Dataset Shape:    (139203, 7)
Full Features:    ['a', 'e', 'i', 'H']
Restrictive Features:  ['a', 'e', 'i']
Target Column:    diameter_log
Model Parameters: {'n_estimators': 150, 'max_depth': 25, 'min_samples_split': 5, 'random_state': 42, 'n_jobs': -1}
================================================

Fold 1/5 Complete | Full R²: 0.8492 | Restricted R²: 0.3319 | Δ R²: 0.5173
Fold 2/5 Complete | Full R²: 0.8514 | Restricted R²: 0.3360 | Δ R²: 0.5154
Fold 3/5 Complete | Full R²: 0.8513 | Restricted R²: 0.3325 | Δ R²: 0.5188
Fold 4/5 Complete | Full R²: 0.8473 | Restricted R²: 0.3321 | Δ R²: 0.5152
Fold 5/5 Complete | Full R²: 0.8549 | Restricted R²: 0.3327 | Δ R²: 0.5222

================================================
CROSS-VALIDATION SUMMARY AVERAGES (RANDOM FOREST)
================================================
Full Model Mean R²:       0.8508
Restricted Model Mean R²: 0.3330
Mean Δ R² (Drop Impact):  0.5178 (± 0.0026)
Full Model Mean MAE:      1.0443 km
Restricted Model Mean MAE:2.4090 km
================================================

====================== HYPOTHESIS TESTING REPORT (RANDOM FOREST) ======================
Testing against Operational Tolerance (δ) = 0.05
Calculated Delta R² (Drop Impact):        = 0.5178
------------------------------------------------------------------------------
Statistical Test: [H1_stat] -> Reject H0_stat: Group 2 features significantly improve out-of-sample R².
Operational Test: [H0_ops] -> Fail to Reject H0_ops: Group 2 feature improvement is > delta (0.05). Restricted subset is insufficient.
==============================================================================
```

Analyze (Models)

```
================================================
INITIALIZING GBR PIPELINE
================================================
Dataset Path:     data/cleaned-dataset-jpl-small-body-asteroids.csv
Dataset Shape:    (139203, 7)
Full Features:    ['a', 'e', 'i', 'H']
Restrictive Features:  ['a', 'e', 'i']
Target Column:    diameter_log
Model Parameters: {'n_estimators': 150, 'max_depth': 8, 'learning_rate': 0.05, 'min_samples_leaf': 2, 'random_state': 42}
================================================

Fold 1/5 Complete | Full R²: 0.8527 | Restricted R²: 0.3582 | Δ R²: 0.4945
Fold 2/5 Complete | Full R²: 0.8547 | Restricted R²: 0.3606 | Δ R²: 0.4941
Fold 3/5 Complete | Full R²: 0.8557 | Restricted R²: 0.3659 | Δ R²: 0.4898
Fold 4/5 Complete | Full R²: 0.8524 | Restricted R²: 0.3652 | Δ R²: 0.4872
Fold 5/5 Complete | Full R²: 0.8584 | Restricted R²: 0.3653 | Δ R²: 0.4931

================================================
CROSS-VALIDATION SUMMARY AVERAGES (GBR)
================================================
Full Model Mean R²:       0.8548
Restricted Model Mean R²: 0.3630
Mean Δ R² (Drop Impact):  0.4917 (± 0.0028)
Full Model Mean MAE:      1.0360 km
Restricted Model Mean MAE:2.3461 km
================================================

====================== HYPOTHESIS TESTING REPORT (GBR) ======================
Testing against Operational Tolerance (δ) = 0.05
Calculated Delta R² (Drop Impact):        = 0.4917
------------------------------------------------------------------------------
Statistical Test: [H1_stat] -> Reject H0_stat: Group 2 features significantly improve out-of-sample R².
Operational Test: [H0_ops] -> Fail to Reject H0_ops: Group 2 feature improvement is > delta (0.05). Restricted subset is insufficient.
==============================================================================
```

Analyze (Models)

```
================================================
INITIALIZING KERAS WIDE_DEEP NETWORK PIPELINE
================================================
Dataset Path:     data/cleaned-dataset-jpl-small-body-asteroids.csv
Dataset Shape:    (139203, 10)
Full Features:    ['a', 'e', 'i', 'H']
Restrictive Features:  ['a', 'e', 'i']
Target Column:    diameter_log
Model Parameters: {'epochs': 100, 'batch_size': 64, 'learning_rate': 0.001, 'patience': 10, 'restore_best_weights': True, 'wide_dim': 1
================================================

Fold 1/5 Complete | Full R²: 0.8006 | Restricted R²: 0.2562 | Δ R²: 0.5444
Fold 2/5 Complete | Full R²: 0.8272 | Restricted R²: -0.0404 | Δ R²: 0.8676
Fold 3/5 Complete | Full R²: 0.8309 | Restricted R²: 0.3083 | Δ R²: 0.5226
Fold 4/5 Complete | Full R²: 0.7944 | Restricted R²: 0.2230 | Δ R²: 0.5715
Fold 5/5 Complete | Full R²: 0.7732 | Restricted R²: 0.2978 | Δ R²: 0.4755

================================================
CROSS-VALIDATION SUMMARY AVERAGES (WIDE_DEEP)
================================================
Full Model Mean R²:       0.8053
Restricted Model Mean R²: 0.2090
Mean Δ R² (Drop Impact):  0.5963 (± 0.1392)
Full Model Mean MAE:      1.2608 km
Restricted Model Mean MAE:4098258727372552614707 2.0000 km
================================================

====================== HYPOTHESIS TESTING REPORT (WIDE_DEEP) ======================
Testing against Operational Tolerance (δ) = 0.0500
Calculated Delta R² (Drop Impact):        = 0.5963
--------------------------------------------------------------------------------
Statistical Test: [H1_stat] -> Reject H0_stat: Group 2 features significantly improve out-of-sample R².
Operational Test: [H0_ops] -> Fail to Reject H0_ops: Group 2 feature improvement is > delta (0.05). Restricted subset is insufficient.
================================================================================
```

Analyze (Models)

```
================================================
INITIALIZING KERAS RESNET NETWORK PIPELINE
================================================
Dataset Path:     data/cleaned-dataset-jpl-small-body-asteroids.csv
Dataset Shape:    (139203, 10)
Full Features:    ['a', 'e', 'i', 'H']
Restrictive Features:  ['a', 'e', 'i']
Target Column:    diameter_log
Model Parameters: {'epochs': 100, 'batch_size': 512, 'learning_rate': 0.0005, 'patience': 10, 'restore_best_weights': True, 'hidden_dim
================================================

Fold 1/5 Complete | Full R²: 0.8166 | Restricted R²: 0.0875 | Δ R²: 0.7290
Fold 2/5 Complete | Full R²: 0.8173 | Restricted R²: 0.2984 | Δ R²: 0.5190
Fold 3/5 Complete | Full R²: 0.8093 | Restricted R²: 0.2519 | Δ R²: 0.5574
Fold 4/5 Complete | Full R²: 0.8103 | Restricted R²: 0.3016 | Δ R²: 0.5087
Fold 5/5 Complete | Full R²: 0.8409 | Restricted R²: 0.3039 | Δ R²: 0.5370

================================================
CROSS-VALIDATION SUMMARY AVERAGES (RESNET)
================================================
Full Model Mean R²:       0.8189
Restricted Model Mean R²: 0.2487
Mean Δ R² (Drop Impact):  0.5702 (± 0.0811)
Full Model Mean MAE:      1.2734 km
Restricted Model Mean MAE:8.3245 km
================================================

====================== HYPOTHESIS TESTING REPORT (RESNET) ======================
Testing against Operational Tolerance (δ) = 0.0500
Calculated Delta R² (Drop Impact):        = 0.5702
--------------------------------------------------------------------------------
Statistical Test: [H1_stat] -> Reject H0_stat: Group 2 features significantly improve out-of-sample R².
Operational Test: [H0_ops] -> Fail to Reject H0_ops: Group 2 feature improvement is > delta (0.05). Restricted subset is insufficient.
================================================================================
```

Analyze (Models)

```
================================================
INITIALIZING KERAS FT_TRANSFORMER NETWORK PIPELINE
================================================
Dataset Path:     data/cleaned-dataset-jpl-small-body-asteroids.csv
Dataset Shape:    (139203, 10)
Full Features:    ['a', 'e', 'i', 'H']
Restrictive Features:  ['a', 'e', 'i']
Target Column:    diameter_log
Model Parameters: {'epochs': 100, 'batch_size': 512, 'learning_rate': 0.0003, 'patience': 10, 'restore_best_weights': True, 'embedding_dim': 16, 'num_heads': 2,
'dropout_rate': 0.05}
================================================

Fold 1/5 Complete | Full R²: 0.8290 | Restricted R²: 0.3032 | Δ R²: 0.5258
Fold 2/5 Complete | Full R²: 0.8308 | Restricted R²: 0.3087 | Δ R²: 0.5221
Fold 3/5 Complete | Full R²: 0.8223 | Restricted R²: 0.3081 | Δ R²: 0.5143
Fold 4/5 Complete | Full R²: 0.8223 | Restricted R²: 0.3161 | Δ R²: 0.5061
Fold 5/5 Complete | Full R²: 0.8358 | Restricted R²: 0.3157 | Δ R²: 0.5202

================================================
CROSS-VALIDATION SUMMARY AVERAGES (FT_TRANSFORMER)
================================================
Full Model Mean R²:       0.8280
Restricted Model Mean R²: 0.3104
Mean Δ R² (Drop Impact):  0.5177 (± 0.0069)
Full Model Mean MAE:      1.1435 km
Restricted Model Mean MAE:2.4215 km
================================================

====================== HYPOTHESIS TESTING REPORT (FT_TRANSFORMER) ======================
Testing against Operational Tolerance (δ) = 0.0500
Calculated Delta R² (Drop Impact):        = 0.5177
------------------------------------------------------------------------
Statistical Test: [H1_stat] -> Reject H0_stat: Group 2 features significantly improve out-of-sample R².
Operational Test: [H0_ops] -> Fail to Reject H0_ops: Group 2 feature improvement is > delta (0.05). Restricted subset is insufficient.
========================================================================
```

Analyze (Models)

```
================================================
INITIALIZING KERAS GATED_RESNET NETWORK PIPELINE
================================================
Dataset Path:     data/cleaned-dataset-jpl-small-body-asteroids.csv
Dataset Shape:    (139203, 10)
Full Features:    ['a', 'e', 'i', 'H']
Restrictive Features:  ['a', 'e', 'i']
Target Column:    diameter_log
Model Parameters: {'epochs': 100, 'batch_size': 512, 'learning_rate': 0.0005, 'patience': 10, 'restore_best_weights': True, 'hidden_dim': 256, 'num_blocks': 3}
================================================


Fold 1/5 Complete | Full R²: 0.8229 | Restricted R²: 0.1872 | Δ R²: 0.6358
Fold 2/5 Complete | Full R²: 0.8185 | Restricted R²: 0.3007 | Δ R²: 0.5178
Fold 3/5 Complete | Full R²: 0.7274 | Restricted R²: 0.0450 | Δ R²: 0.6824
Fold 4/5 Complete | Full R²: 0.7397 | Restricted R²: -0.0532 | Δ R²: 0.7929
Fold 5/5 Complete | Full R²: 0.8351 | Restricted R²: 0.2660 | Δ R²: 0.5691

================================================
CROSS-VALIDATION SUMMARY AVERAGES (GATED_RESNET)
================================================
Full Model Mean R²:       0.7887
Restricted Model Mean R²: 0.1492
Mean Δ R² (Drop Impact):  0.6396 (± 0.0950)
Full Model Mean MAE:      4.4961 km
Restricted Model Mean MAE:2554938455683444506624.0000 km
================================================

====================== HYPOTHESIS TESTING REPORT (GATED_RESNET) ======================
Testing against Operational Tolerance (δ) = 0.0500
Calculated Delta R² (Drop Impact):        = 0.6396
----------------------------------------------------------------------
Statistical Test: [H1_stat] -> Reject H0_stat: Group 2 features significantly improve out-of-sample R².
Operational Test: [H0_ops] -> Fail to Reject H0_ops: Group 2 feature improvement is > delta (0.05). Restricted subset is insufficient.
======================================================================
```

Analyze (Models)

The models were trained and tested against the hypotheses, quantified as:

$$\Delta R^2_{test} = R^2_{Model(\mathbf{x}full)} - R^2_{Model(\mathbf{x}restricted)}$$

ΔR^2_{test} was used to reject or fail to reject the null hypotheses. Every model resulted in ΔR^2_{test} greater than the statistical and operational threshold (= 0.05). This means that the restricted features subset was not sufficient to accurately predict the diameter of an asteroid for operational use. As such, final model determination was based on the greatest full feature subset model mean R^2. While all models met or exceeded a mean R^2 of 0.7887, it was the traditional models of Random Forest and Gradient Boosting Regression that scored the best with a mean R^2 of 0.8508 and 0.8548, respectively. The determination of which model will be used in production will undergo an analysis of average Mean Absolute Percentage Error (MAPE) to determine how well predictions matched actual values of asteroids. The model with the lowest average MAPE will be selected for use in production.

Analyze (Models)

A subset of asteroids diameters were predicted with each model (full and restricted trained models) ranging from Sub-Kilometer, Medium, and Massive to validate accuracy and the results logged for analysis.


```
asteroids = [
    {"spkid": 20000001, "name": "Ceres", "actual": 939.4, "data": {"a": 2.766, "e": 0.0797, "i": 10.59, "H": 3.34}},
    {"spkid": 20020000, "name": "Varuna", "actual": 900.0, "data": {"a": 43.2, "e": 0.0516, "i": 17.14, "H": 3.79}},
    {"spkid": 20000004, "name": "Vesta", "actual": 522.77, "data": {"a": 2.361, "e": 0.0902, "i": 7.14, "H": 3.25}},
    {"spkid": 20000002, "name": "Pallas", "actual": 513.0, "data": {"a": 2.77, "e": 0.2307, "i": 34.93, "H": 4.12}},
    {"spkid": 20000003, "name": "Juno", "actual": 246.596, "data": {"a": 2.671, "e": 0.2557, "i": 12.99, "H": 5.19}},
    {"spkid": 20000065, "name": "Cybele", "actual": 237.26, "data": {"a": 3.407, "e": 0.1279, "i": 3.57, "H": 6.83}},
    {"spkid": 20000560, "name": "Delila", "actual": 35.334, "data": {"a": 2.753, "e": 0.1579, "i": 8.45, "H": 10.94}},
    {"spkid": 20001203, "name": "Nanna", "actual": 35.18, "data": {"a": 2.882, "e": 0.2515, "i": 6.0, "H": 11.65}},
    {"spkid": 20001815, "name": "Beethoven", "actual": 30.598, "data": {"a": 3.149, "e": 0.1932, "i": 2.74, "H": 11.39}},
    {"spkid": 20004864, "name": "Nimoy", "actual": 11.874, "data": {"a": 2.468, "e": 0.1782, "i": 3.62, "H": 13.6}},
    {"spkid": 20012358, "name": "Azzurra", "actual": 11.75, "data": {"a": 2.79, "e": 0.1143, "i": 7.61, "H": 13.74}},
    {"spkid": 20013493, "name": "Lockwood", "actual": 11.101, "data": {"a": 2.668, "e": 0.1995, "i": 12.91, "H": 13.77}},
    {"spkid": 20005855, "name": "Yukitsuna", "actual": 11.1, "data": {"a": 2.555, "e": 0.154, "i": 15.5, "H": 12.13}},
    {"spkid": 20013123, "name": "Tyson", "actual": 10.875, "data": {"a": 2.359, "e": 0.2725, "i": 23.27, "H": 12.63}},
    {"spkid": 20001946, "name": "Walraven", "actual": 9.205, "data": {"a": 2.294, "e": 0.2345, "i": 8.16, "H": 12.27}},
    {"spkid": 20000970, "name": "Primula", "actual": 9.204, "data": {"a": 2.559, "e": 0.2723, "i": 5.03, "H": 12.25}},
    {"spkid": 20002445, "name": "Blazhko", "actual": 9.13, "data": {"a": 2.269, "e": 0.1467, "i": 6.07, "H": 12.39}},
    {"spkid": 20008626, "name": "Melissarauch", "actual": 4.266, "data": {"a": 3.169, "e": 0.1038, "i": 6.18, "H": 14.46}},
    {"spkid": 20012503, "name": "Danielpeterson", "actual": 4.266, "data": {"a": 2.756, "e": 0.1282, "i": 8.75, "H": 14.36}},
    {"spkid": 20003072, "name": "Vilnius", "actual": 4.26, "data": {"a": 2.239, "e": 0.1799, "i": 5.65, "H": 13.68}},
    {"spkid": 20091287, "name": "Simon-Garfunkel", "actual": 1.745, "data": {"a": 2.289, "e": 0.1076, "i": 6.23, "H": 15.83}},
    {"spkid": 20001865, "name": "Cerberus", "actual": 1.2, "data": {"a": 1.08, "e": 0.4669, "i": 16.1, "H": 16.77}},
    {"spkid": 20001915, "name": "Quetzalcoatl", "actual": 0.5, "data": {"a": 2.545, "e": 0.5703, "i": 20.4, "H": 18.38}},
    {"spkid": 20003757, "name": "Anagolay", "actual": 0.5, "data": {"a": 1.835, "e": 0.4458, "i": 3.87, "H": 19.01}},
    {"spkid": 20101955, "name": "Bennu", "actual": 0.48444, "data": {"a": 1.126, "e": 0.2037, "i": 6.03, "H": 20.21}},
    {"spkid": 20398188, "name": "Agni", "actual": 0.462, "data": {"a": 0.8646, "e": 0.2734, "i": 13.25, "H": 19.33}},
    {"spkid": 20004660, "name": "Nereus", "actual": 0.33, "data": {"a": 1.485, "e": 0.3587, "i": 1.45, "H": 18.75}}
]
```

Analyze (Models)

```
================================================================================
MAPE & VOLATILITY MATRIX WITH MODEL
================================================================================
```

		Full: Avg MAPE	Full: Error Volatility (Std)	Restricted: Avg MAPE	Restricted: Error Volatility (Std)
keras_ft_transformer	Massive (>50km)	52.24%	23.14%	98.51%	1.17%
	Medium (1-50km)	24.72%	17.86%	63.60%	21.15%
	Sub-Kilometer (<1km)	44.13%	47.00%	60.16%	50.76%
	Average	34.43%	27.86%	70.72%	29.76%
keras_gated_resnet	Massive (>50km)	326.62%	700.98%	98.86%	0.86%
	Medium (1-50km)	30.10%	18.53%	59.85%	25.12%
	Sub-Kilometer (<1km)	83.41%	77.87%	224.78%	68.57%
	Average	105.87%	332.42%	99.06%	71.22%
keras_resnet	Massive (>50km)	19.07%	7.24%	91.34%	18.58%
	Medium (1-50km)	25.81%	18.43%	64.81%	22.69%
	Sub-Kilometer (<1km)	69.42%	64.14%	59.46%	73.57%
	Average	32.39%	34.21%	69.71%	36.59%
keras_wide_deep	Massive (>50km)	21.52%	20.06%	83.56%	37.26%
	Medium (1-50km)	40.31%	26.15%	62.89%	22.91%
	Sub-Kilometer (<1km)	46.88%	40.10%	82.55%	109.68%
	Average	37.35%	28.28%	71.13%	50.23%
rf	Massive (>50km)	18.03%	12.48%	86.25%	12.45%
	Medium (1-50km)	11.24%	9.32%	41.49%	19.02%
	Sub-Kilometer (<1km)	20.37%	20.39%	30.84%	21.48%
	Average	14.44%	12.65%	49.47%	26.97%
xgboost	Massive (>50km)	8.10%	4.20%	85.92%	24.31%
	Medium (1-50km)	20.81%	17.78%	59.50%	22.07%
	Sub-Kilometer (<1km)	35.98%	30.47%	27.04%	22.17%
	Average	20.80%	20.25%	59.36%	28.88%

Analyze (Models)

The Mean Absolute Percentage Error (MAPE) and volatility (standard deviation) matrix was used to see how well each model predicted the size of an asteroid. It was broken down into categories by asteroid sizes (Massive, Medium, and Sub-Kilometer) to see how well it managed the skewed data. The results showed that the classical models had the best average among all categories with Random Forest having a MAPE of 14.44% and GBR (as xgboost in code) having a 20.79% for the full feature model. This means the top classical models deviate from the actual diameter by an average of only 15% to 21% across all samples.

In general, each model struggled with the skewed data specifically with the Sub-Kilometer categories. The MAPE score for all models in this category skyrocked from the RF at 20.37% up to GRN at 83.41%. This makes sense as there are many more smaller asteroids and smaller asteroids tend to be harder to gather data on.

Most notable is the complete collapse of all accuracy in predicting diameter of an asteroid in the restricted feature subset. MAPE for all models skyrockted from an average of 49.47% with RF up to 99.06% with GRN. This suggests that absolute magnitude is highly correlated to diameter and orbital characterisitcs alone can not reliably predicit the size of an asteroid without H.

Analyze (Models)

MUTUAL INFORMATION GEOMETRIC ANCHOR REPORT:

==========

Feature 'H': Mutual Information Score = 0.7674

Feature 'a': Mutual Information Score = 0.2201

Feature 'e': Mutual Information Score = 0.0488

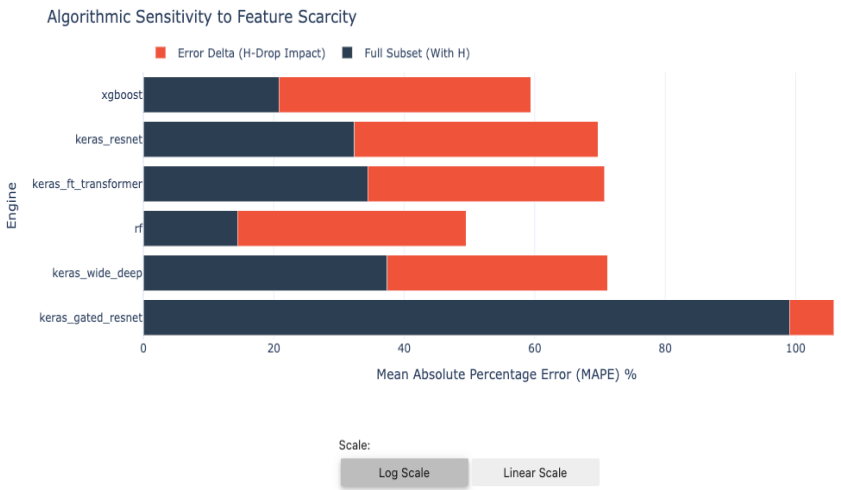
Feature 'i': Mutual Information Score = 0.0269

==========

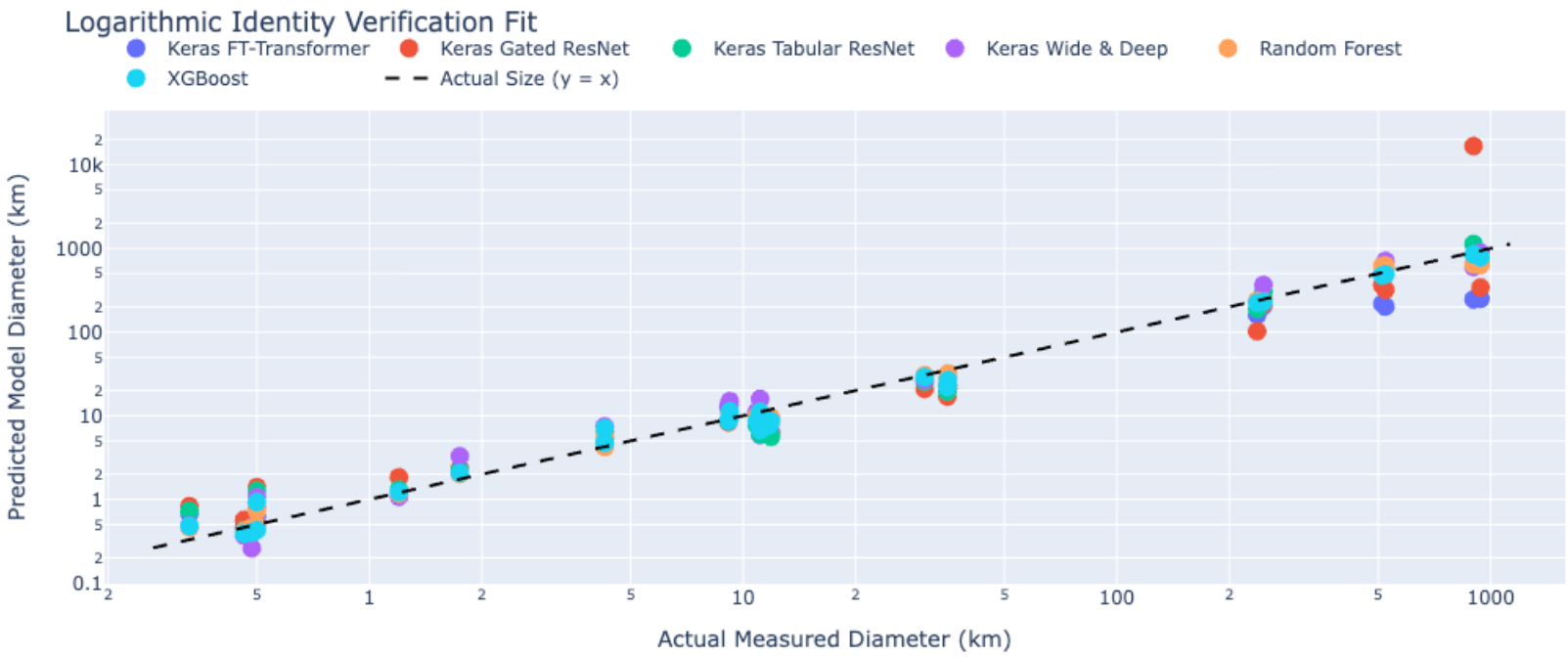
Looking at the regression of diameter to subset features, it supports the MAPE results. H accounts for 76.74% of the predicting power for diameter. The remaining features have low, if any, direct correlation to diameter.

Analyze (Models)

Reviewing the sensitivity figure, it is clear that H is necessary to predict diameter. Error rates do uble when H is removed as can be seen by RF which had a MAP E increase from 14.44% to 49.47 % when H was removed.



Analyze (Models)

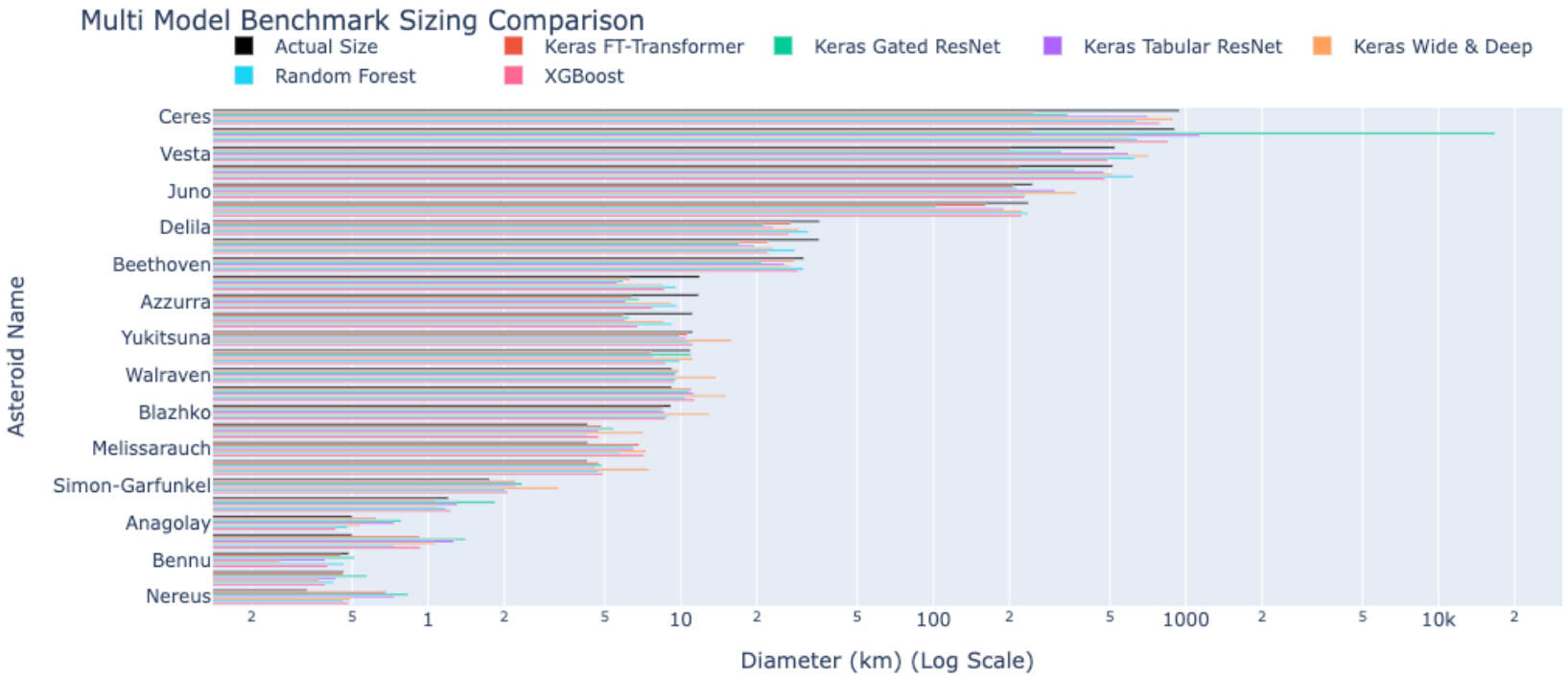


Analyze (Models)

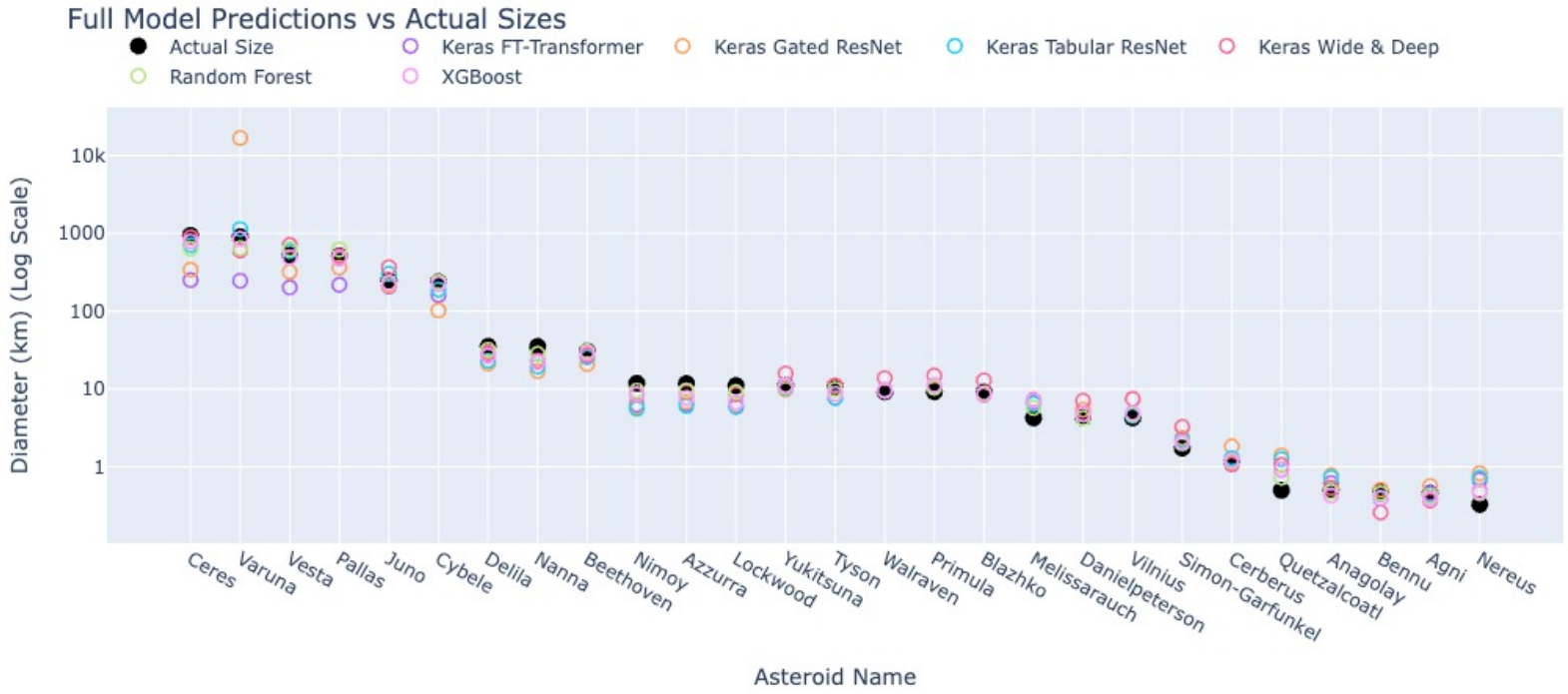
The linear-to-logarithmic visualization of predicted versus actual sizes confirms that the dataset is heavily skewed, validating the operational necessity of the target logarithmic transformation. The figure demonstrates that while all six architectures successfully handled predictions within the Medium range, they struggled when handling outliers in the Massive range.

Classical models like Random Forest (RF) handled the scale boundaries the most effectively, based on how tightly their predictions cluster along the ideal identity fit line. Conversely, the Wide & Deep (W&D) network performed relatively well with Massive outliers but degraded severely when predicting in the Sub-Kilometer range. Because W&D architectures are structurally optimized to capture broad, population-wide averages, the combination of extreme dataset skewness and the physical inconsistency of small asteroids to reliably reflect light prevented the linear wide path from establishing effective rules to reliably predict diameter.

Analyze (Models)



Analyze (Models)



Analyze (Models)

The model benchmark figure supports that the classical models like RF and GBR perform better than the neural network models. All models appear to overpredict on Sub-Kilometer asteroids and underpredict on Massive asteroids. The tree-based models struggle with predicting larger asteroids because they are unable to predict diameters that exceed what they received during training. This can be seen in RF where Ceres and Pallas were predicted to be similar in size on the upper outlier even though Ceres is hundred of kilometers larger. This also supports tree-based models better accuracy because they are step-based and just split data base on rules. Neural networks tend to want to fit data to a pattern. Since the dataset is highly skewed and does not have a linear or reliable pattern, it makes tree-based models more accurate at predicting diameter.

Report & Act

1. Hypothesis Testing

Acceptance of Statistical Alternative Hypothesis ($H_{1,stat}$): $\Delta R^2_{test} > 0$

- *Definition:* The inclusion of Group 2 features yields an increase in the out-of-sample ΔR^2_{test}, indicating that the expanded features **do** increase the accuracy of predicting the target diameter (Y).
- *Result:* All six models resulted in a ΔR^2_{test} of no less than 0.4917. This exceeded the statistical threshold of zero (0). This means the expanded features did increase the accuracy of predicting the target diameter (Y).

Acceptance of Operational Null Hypothesis ($H_{0,ops}$): $\Delta R^2_{test} > \delta$

- *Definition:* The inclusion of Group 2 features yield an operationally significant increase in accuracy to predict target diameter (Y), indicating that the restricted feature subset **is not** sufficient enough for operational use.
- *Result:* All six models resulted in a ΔR^2_{test} of no less than 0.4917. This exceeded the operational threshold of 0.05. This means the restricted feature subset is insufficient for production deployment.

Report & Act

2. Feature Variance Analysis

- Feature importance and sensitivity metrics confirm that absolute magnitude (H) accounts for 76.74% of the variance explained when predicting diameter. Omitting H from the training pipeline more than triples the error rate across all models, demonstrating that the geometric elements (a, e, i) lack sufficient standalone signal and H is required to reliably and accurately predict diameter.

Report & Act

3. Model Performance & Selection

- **Final Model Selection:** Random Forest (RF)
  - RF is formally selected as the most reliable deployment model for predicting asteroid diameter. Across five-fold cross-validation on the full feature set, RF achieved a Mean R^2 of 0.8508 and an out-of-sample Mean MAE of 1.0443 km. While Gradient Boosting Regression (GBR, XGBOOST in code) demonstrated a slightly higher Mean R^2 of 0.8548 and a comparable Mean MAE of 1.0360 km, RF was ultimately selected for production deployment because it resulted in the lowest overall average Mean Absolute Percentage Error (MAPE) of 14.44% compared to the average MAPE for GBR of 20.80%.

Report & Act

4. Classical Modeling Insights

- Classical tree-based machine learning models performed the best for highly skewed asteroid data. This is likely because it utilizes discrete, step-based data partitioning to isolate numeric subgroups for predicting diameter. Tree-based models are limited because they can only predict within the range of their training data but were able to establish more reliable patterns in that range for predicting diameter. For instance, while GBR handled extreme values better at the Massive (>50km) scale (8.10% vs 18.03% MAPE), RF outperformed it at the highly populous Sub-Kilometer (<1km) scale (20.37% vs 35.98% MAPE), securing the lowest global error average.

Report & Act

5. Neural Network Insights

- Neural networks struggled to reliably predict diameter because they favor optimizing for generalized pattern fitting. The highly skewed dataset and inconsistent measurements of features like H were likely over-smoothed. This caused the models to overpredict at the Sub-Kilometer scale and struggle to handle the Massive scale. Furthermore, removing H caused complete mathematical failures during training. Without this feature, both the W&D and GRN models completely broke down on certain asteroid scales, dropping to negative accuracy levels and throwing massive errors exceeding 10^{21} kilometers.

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